

responsibilities.

4. will become effective communicators in professional and non-professional environments and be able to function as a team member.

The EE program has the following student outcomes:

- Outcome 1: An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
- Outcome 2: An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
- Outcome 3: An ability to communicate effectively with a range of audiences.
- Outcome 4: An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
- Outcome 5: An ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives.
- Outcome 6: An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
- Outcome 7: An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

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Program of Study: Major: Mathematics

Program Overview

The Bachelor of Science in Mathematics provides students with a strong applied and theoretical foundation in mathematics that enables them to pursue either industry employment or graduate studies. Students in the program gain experience using a variety of modeling techniques in combination with technology to solve real-world problems as well as develop a deep understanding of the generalizations and rigor that mathematics has to offer.

Mathematics majors are supported to pursue opportunities for learning outside of the classroom such as undergraduate research with a professor, external Research Experience

for Undergraduates (REU) programs, or an internship with a local industry partner.

This program of study leads to the following credential:

- Bachelor of Science degree with a major in Mathematics

Bachelor of Science degree with a major in Mathematics

Credential Overview

Mathematics is an established and growing field with students pursuing careers in engineering, actuarial science, database and computer systems administration network and data communication analysis, statistical analysis secondary mathematics teaching and other fields. Students with a major in mathematics often pursue graduate studies in mathematics, physics, and engineering.

Completion Requirements

Calculus Requirements (15 Credits)

Complete the following courses with a minimum 2.50 cumulative GPA, and no grade below 2.0 in each course.

- STMATH 124 – Calculus I
- STMATH 125 – Calculus II
- STMATH 126 – Calculus III

Core Requirements (45 credits)

Complete the following courses with a minimum grade of 2.0 in each course:

- STMATH 207 Differential Equations
- STMATH 208 Matrix Algebra
- STMATH 224 Multivariable Calculus
- STMATH 300 Foundations of Modern Mathematics
- STMATH 301 Introduction to Professional Mathematics
- STMATH 402 Abstract Algebra I

- STMATH 424 Introduction to Analysis I
- STMATH 341 Introduction to Statistical Inference, STMATH 390 Probability and Statistics in Engineering, or STMATH 392 Probability
- STMATH 381 Discrete Mathematical Modeling or STMATH 405 Numerical Analysis I

Mathematics Electives (25 credits)

Complete five additional STMATH courses at the 300- or 400-level with a minimum grade of 2.0 in each course, distributed as follows:

- (15 credits) At least three elective courses chosen from the following list, with at least one of these courses having an asterisk (*) preceding it.
 - *STMATH 403 Abstract Algebra II
 - *STMATH 406 Numerical Analysis II
 - *STMATH 408 Nonlinear Optimization
 - *STMATH 409 Advanced Linear Algebra
 - *STMATH 425 Real Analysis II
 - *STMATH 441 Topology
 - STMATH 407 Linear Programming
 - STMATH 420 History of Mathematics
 - STMATH 427 Complex Analysis
 - STMATH 444 Foundations of Geometry
- (10 credits) Two additional elective courses chosen from a program-approved list of electives.
- No more than five credits of STMATH 498 or STMATH 499 may count toward elective credits.

Computer Programming Requirement (5–6 credits)

A minimum grade of 2.0 in both courses from one of the following options:

- CSS 123 and CSSSKL 123
- CSS 132 and CSSSKL 132

- CSS 142 and CSSSKL 142

Additional Courses

As needed to fulfill University General Education Requirements and to equal 180 credits.

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Additional Information

Current research interests of the UWB mathematics faculty include: computed tomography; discrete geometry and tiling theory; knot theory; mathematics education; matrix theory, matrix analysis, and the geometry of polynomials; and numerical optimization, nonlinear optimization, and non-smooth optimization.

Employers often state that they prefer the critical thinking skills acquired in the mathematics major paired with secondary skills obtained from a strategically selected minor in another discipline instead of vice versa. Some strategically selected minors include, but are not limited to: Actuarial Science, Computer Science & Software Engineering, Economics, Physics, Business Administration, Education and Society, and Teaching and Learning. Pre-med and pre-law students majoring in mathematics have historically scored higher on the MCAT and LSAT exams than other majors. Careers in mathematics are frequently rated among the best jobs.

Preparation for a career in teaching mathematics

The curriculum represents a standard mathematics degree; however the courses offered also reflect the recommendations put forth by the Mathematical Association of America's (MAA) CUPM Curriculum Guide 2004 (reference is Mathematical Association of America (2004). Undergraduate Programs and Courses in the Mathematical Sciences: CUPM Curriculum Guide 2004. Ret May 2011: http://www.maa.org/cupm/curr_guide.html) for majors preparing to be secondary mathematics teachers. According to MAA, mathematical sciences majors preparing to teach secondary mathematics should:

- Learn to make appropriate connections between the advanced mathematics they are learning and the secondary mathematics they will be teaching. They should be helped to reach this understanding in courses throughout the curriculum and through a senior-level experience that makes these connections explicit.
- Fulfill the requirements for a mathematics major by including topics from abstract algebra and number theory, analysis (advanced calculus or real analysis), discrete mathematics, geometry, and statistics and probability with an emphasis on data analysis;
- Learn about the history of mathematics and its applications, including recent work;
- Experience many forms of mathematical modeling and a variety of technological tools, including graphing calculators and geometry software.

Links:

- Mathematical Association of America (MAA) <http://www.maa.org/>
- National Council of Teachers of Mathematics <http://www.nctm.org/>

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Program of Study: Major: Mechanical Engineering

Program Overview

As an ABET accredited engineering program, the Bachelor of Science in Mechanical Engineering curriculum emphasizes hands-on experience, collaborative problem solving, and societal implications in the design, production, and implementation of mechanical and thermal fluid systems. It also complements the existing Bachelor of Science in Electrical Engineering major by providing additional learning and research opportunities in biomedical engineering and in power engineering, where electrical and mechanical technologies interweave.

Prerequisites may be met through coursework at UW Bothell, another four-year institution, or a community college. Graduates will be prepared for a wide variety of careers inside and outside of engineering, or for continuation of study at the graduate level.

This program of study leads to the following credential:

- Bachelor of Science in Mechanical Engineering degree

Admission Requirements